Design Documentation

Interactive House – SG2: Units

Revision History

Date	Version	Description	Author
25/Feb/26	1.0	Initial Design + Diagrams for Web (D1–D4)	Mustafa Al-Bayati
25/Feb/26	1.1	Initial Design + Diagrams for Mobile (D5–D7)	Daniel Jönsson

Design item List

Requirement Name	Priority
D1. Web Welcome Screen	Desirable
D2. Web User Authentication (Login/Signup)	Essential
D3. Web Device Management Interface	Essential
D4. Web Hub Control Panel	Essential
D5. Mobile Login/Signup	Essential
D6. Mobile Control Devices in Device Hub	Essential
D7. Mobile Database Connection	Essential

Design Item Descriptions

D1. Web Welcome Screen (Desirable)

Description

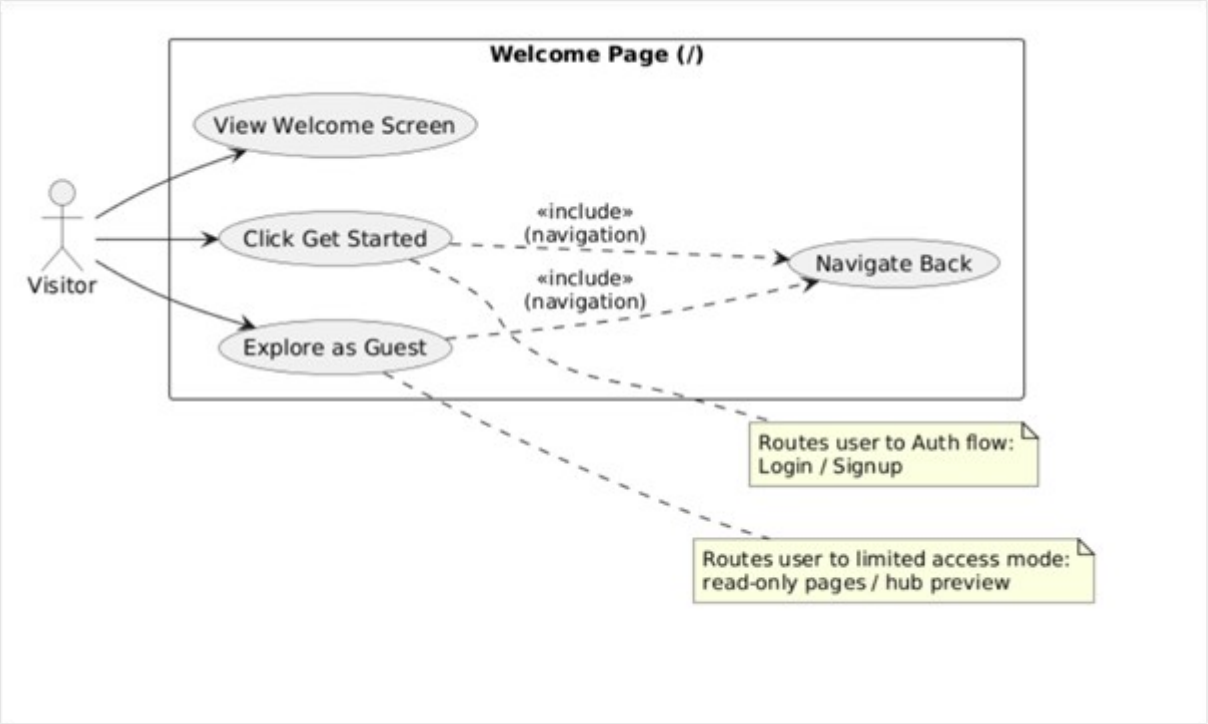
First screen of the web application showing Interactive House branding and entry options.

Components

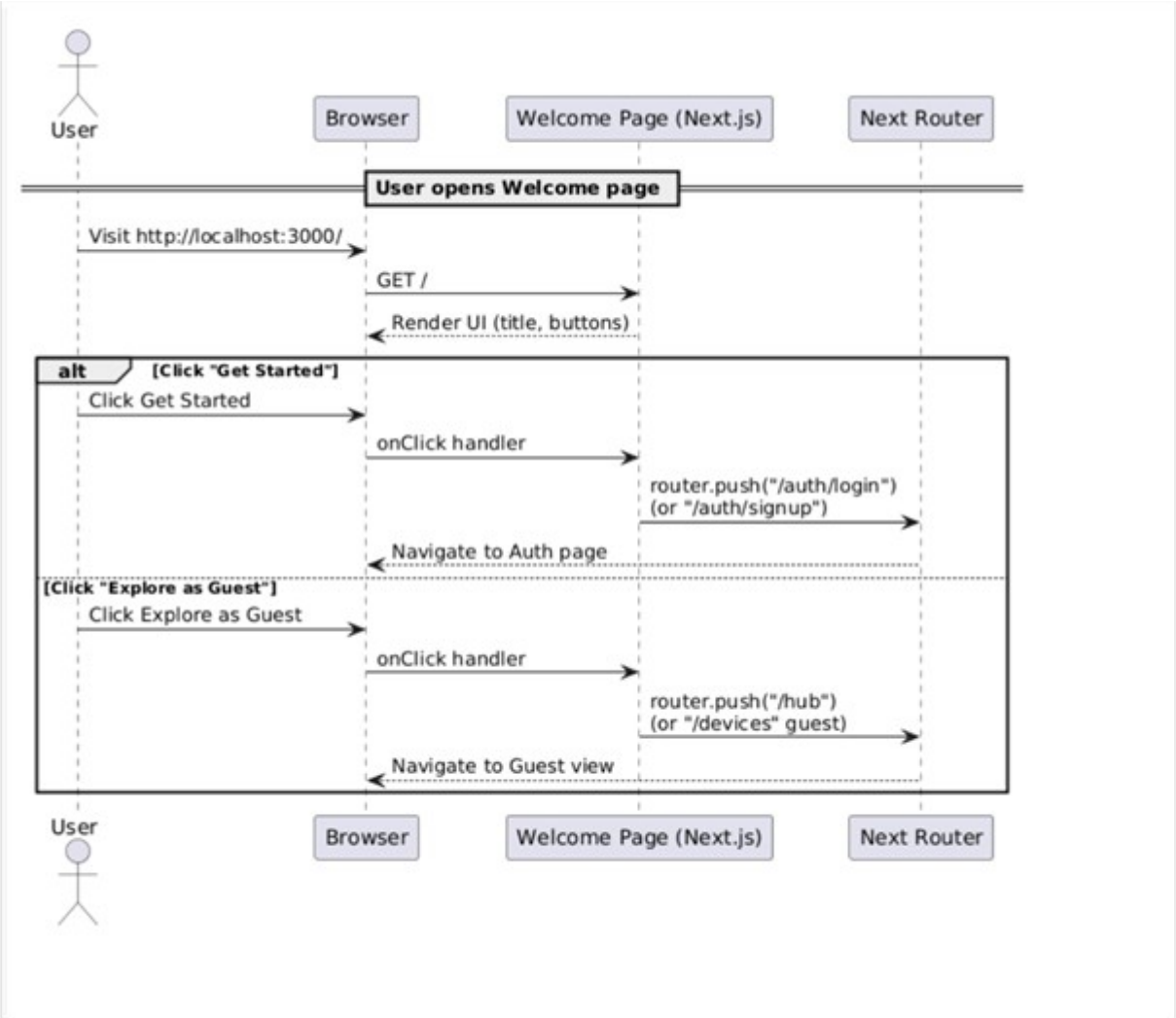
App logo + title + short tagline
Get Started button
Explore as Guest link (limited access)

Design

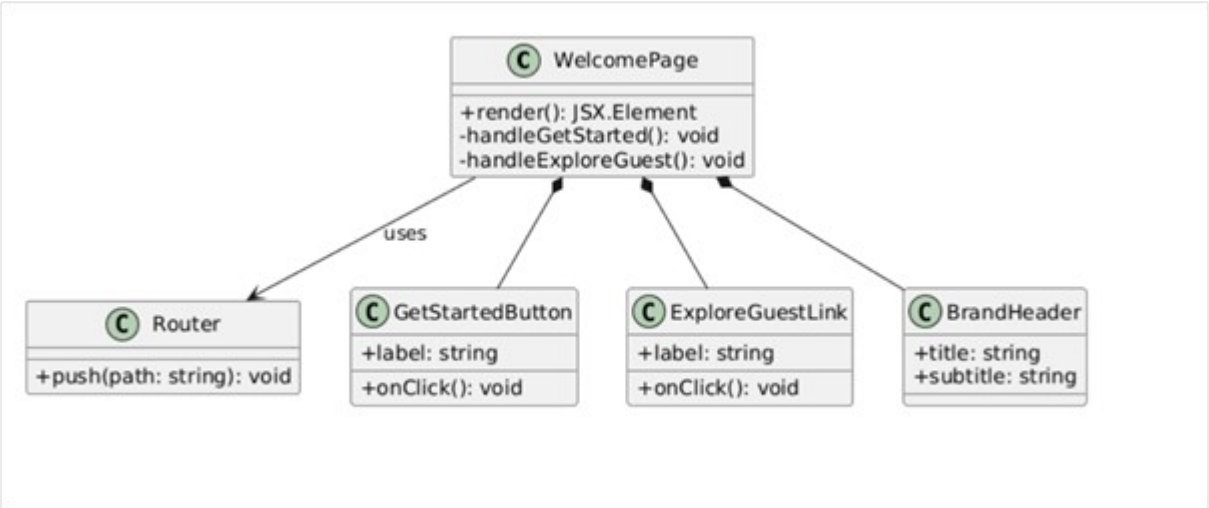
Get Started routes the user to Login/Signup.
Guest mode allows viewing pages (e.g., Devices list read-only) but blocks actions like toggling actuators and database write actions.



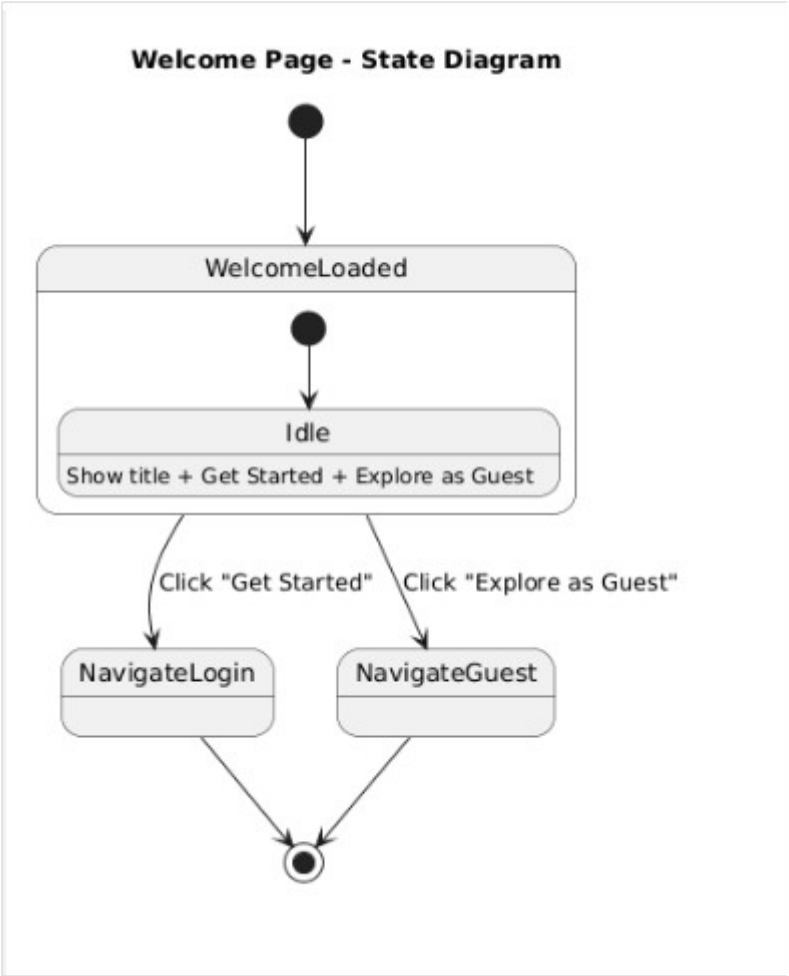
Design item 1 Use Case Diagram, Welcome Page



Design item 2 Sequence Diagram, Welcome Page Navigation



Design item 3 Class Diagram, Welcome Page Structure (UI + Navigation)



Design Item 4 Welcome page, State Diagram

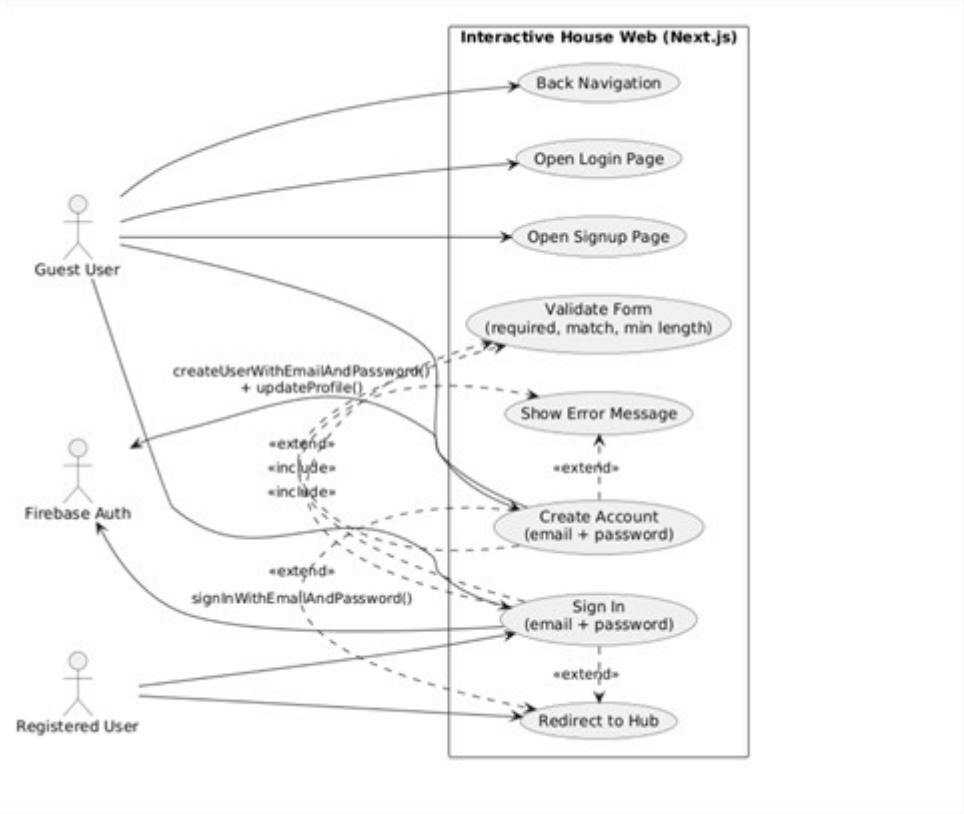
D2. Web User Authentication (Login/Signup) (Essential)

Description
Secure login and signup for registered users to access full system features.

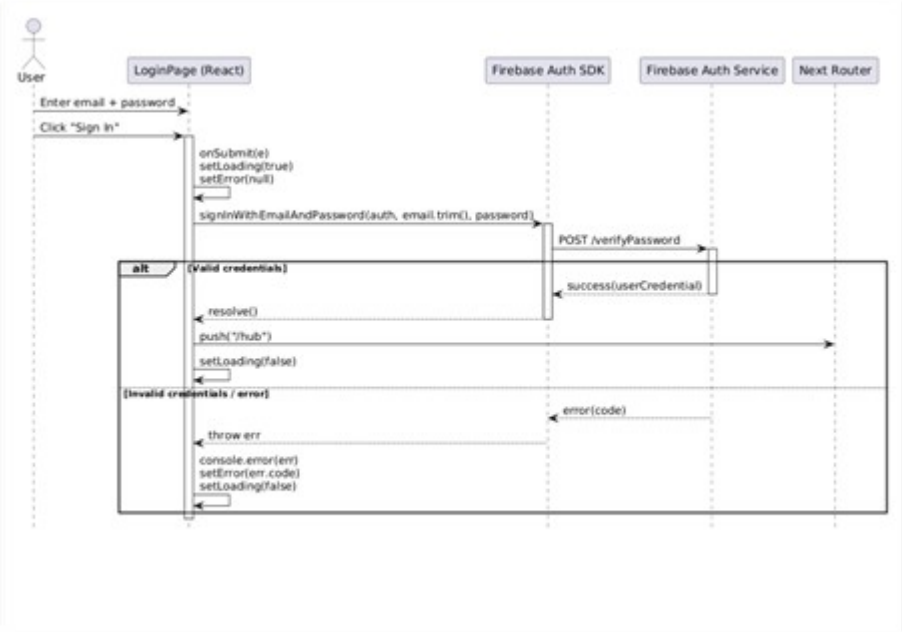
Components
Email & Password Authentication
Sign In button
Signup link

Design
Users must authenticate before performing any control actions.
Passwords are never stored in plain text; only password hashes.
After successful login: redirect to Devices or Hub screen.
Security constraints:

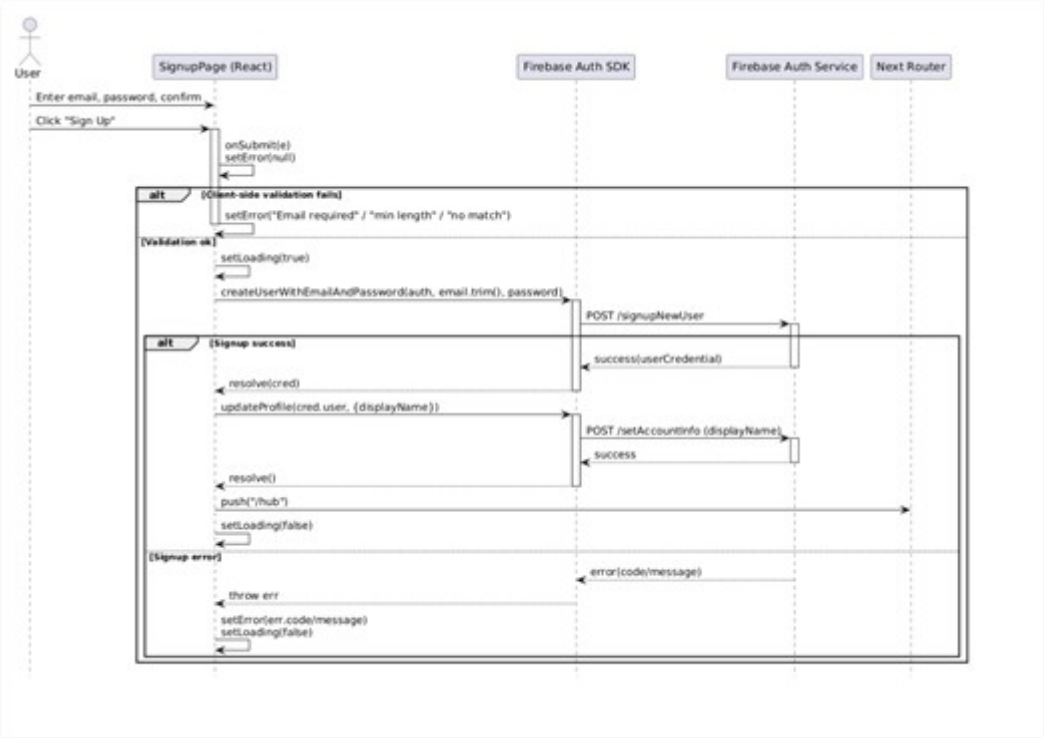
- Minimum password length (e.g., 8)
- Basic rate limiting on login attempts (server-side)



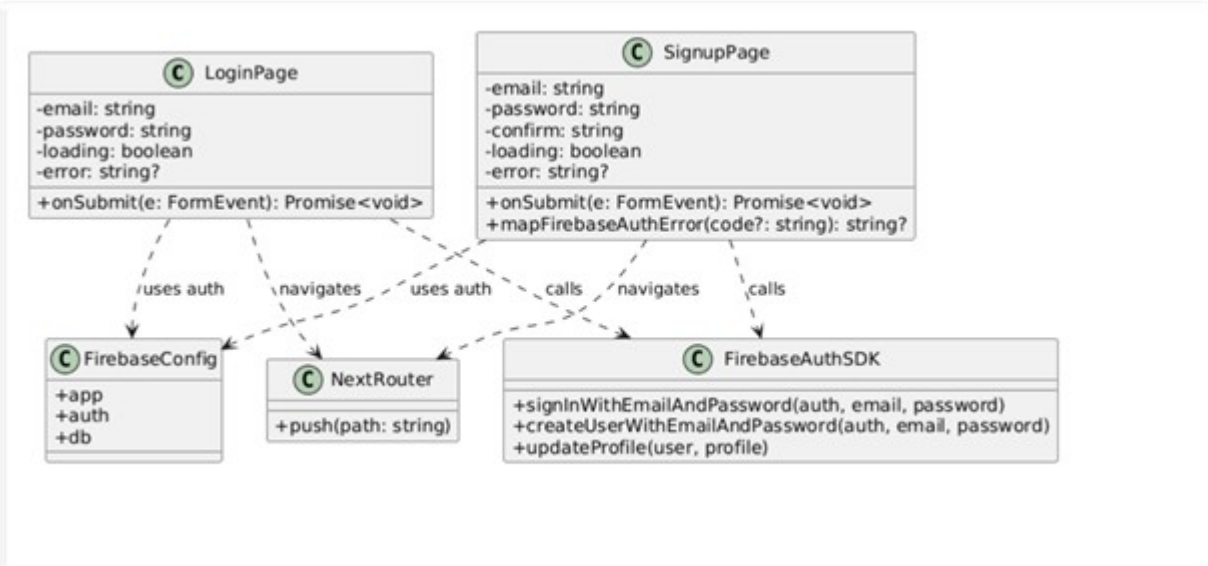
Design item 1 Login/Signup, Use Case Diagram



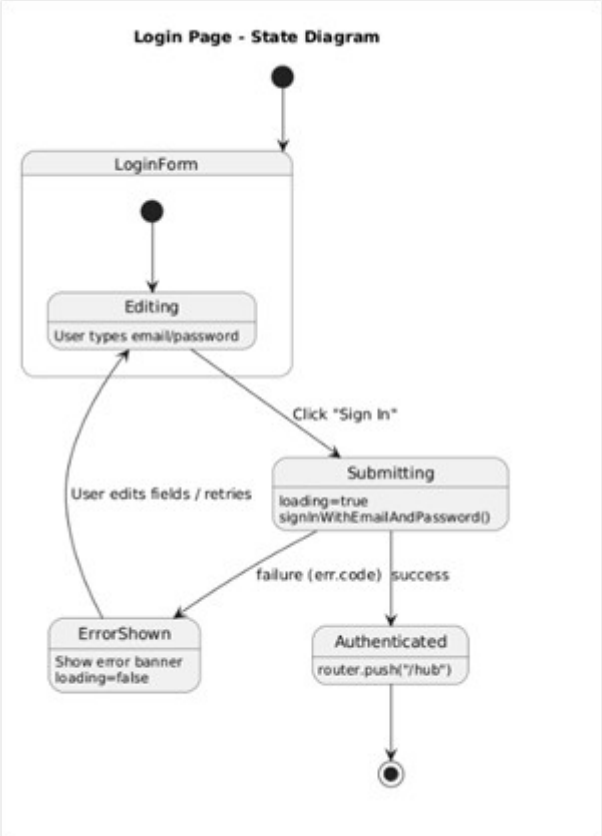
Design Item 2 Login, Sequence Diagram



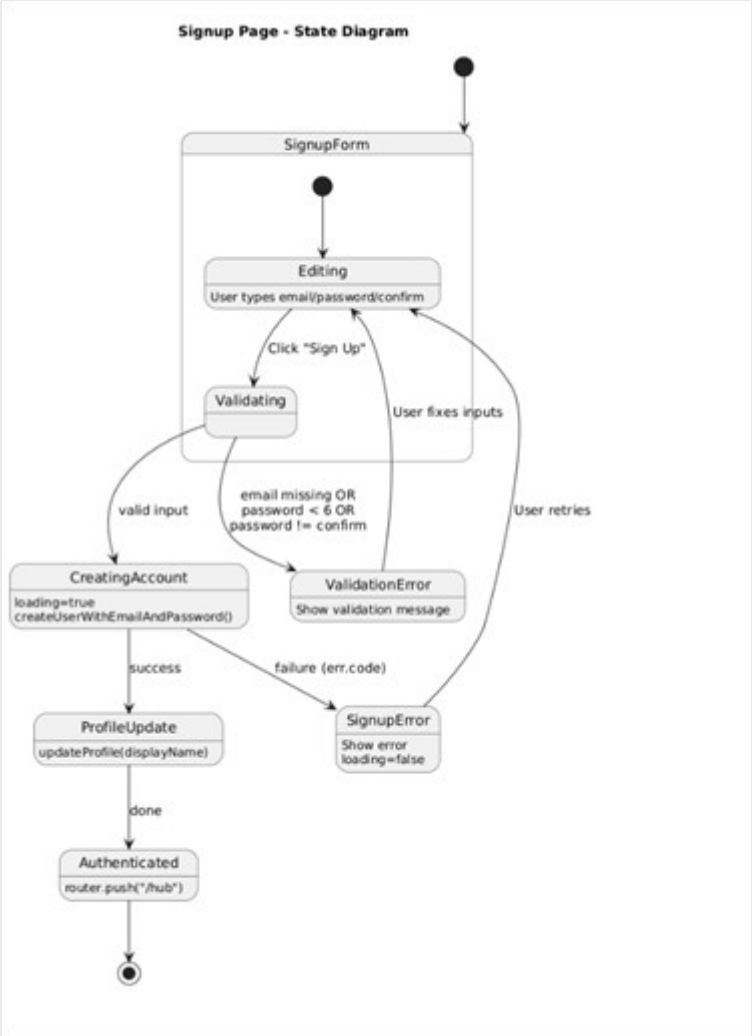
Design Item 3 Signup, Sequence Diagram



Design Item 4 Login/Signup, Class Diagram



Design Item 5 Login Page, State Diagram



Design Item 6 Signup Page, State Diagram

D3. Web Device Management Interface (Essential)

Description

A devices overview page listing all connected devices/sensors with type, pin, and status.

Components

Device cards grid (2 columns)

Each card shows:

- Device icon
- Device name (e.g., Servo 1 (Door))
- Pin label (e.g., D9, D11/A0, I2C SDA/SCL)

Top navigation tabs: Devices, AI, Speech, Hub, Database

Design

Devices" page is read-only in Guest mode, interactive for logged-in users.

Cards are consistent with dark UI + rounded corners + soft shadows (like screenshots).

Clicking a card can route to a detail/control page (optional extension); in this design, most controls are centralized in Hub (D4/D5).

D4. Web Hub Control Panel (Essential)

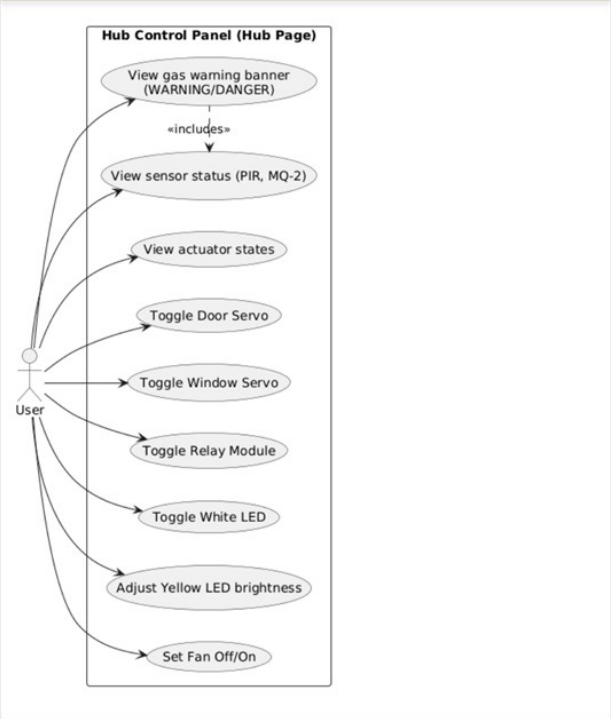
Description
Main control dashboard to control actuators and monitor sensors.

Components
Actuators section:
- Servo toggles (Door/Window)
- Relay toggle
- White LED toggle

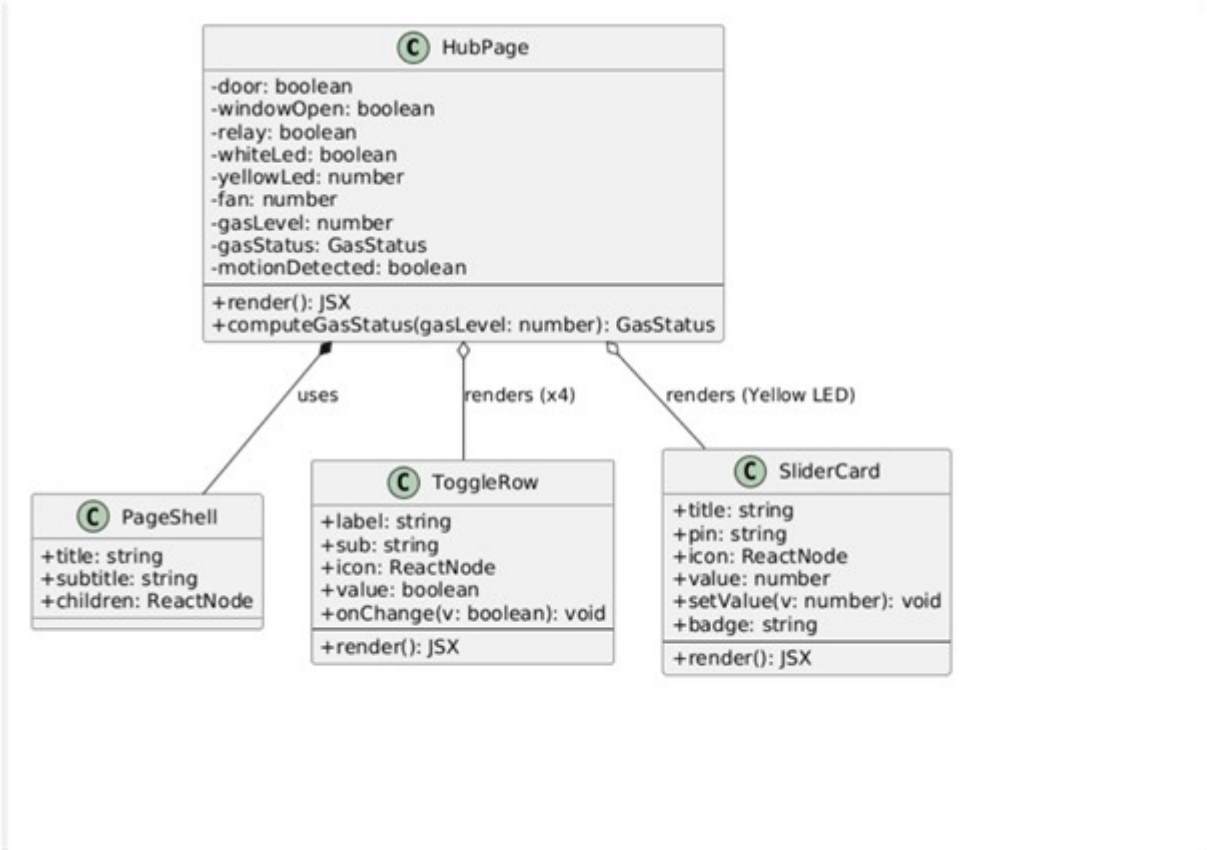
Sensors section:
- PIR status (motion detected / not)

UI widgets:
- Switch toggles
- Slider for LED brightness (Yellow LED)
- Slider for Fan speed
- Fan mode segmented control (Off/On + “Normal” mode badge)

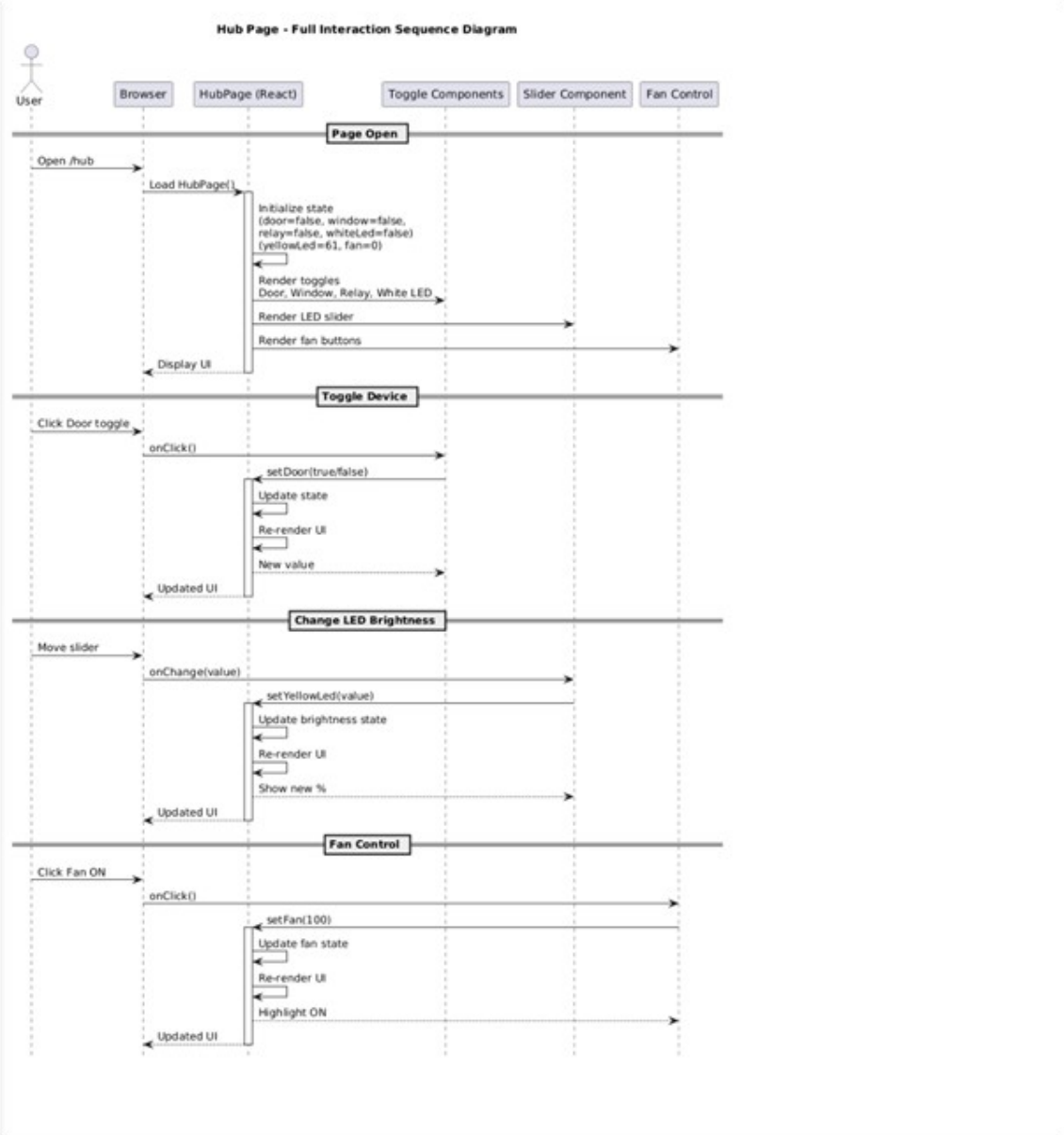
Design
Hub is the primary place for controlling actuators.
UI updates must reflect real device state



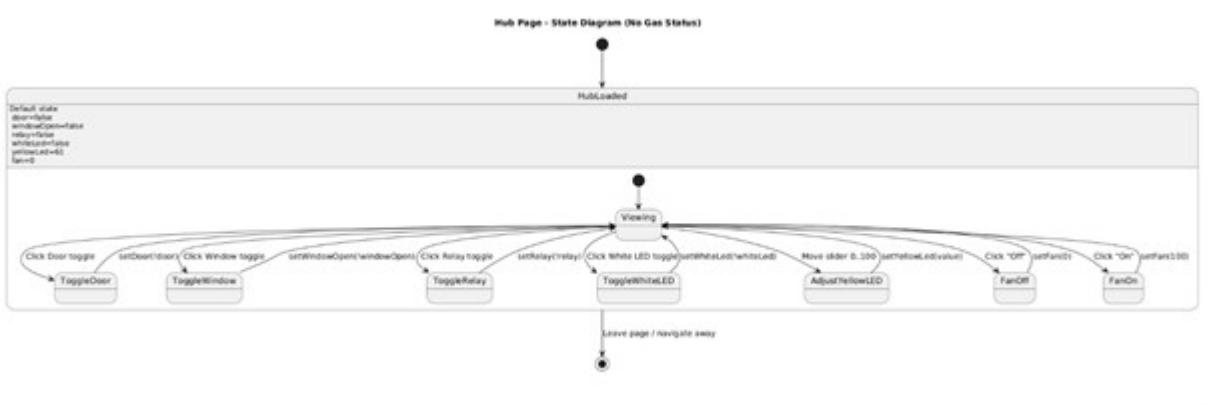
Design Item 1 Use Case Diagram, Hub Page



Design Item 2 Class Diagram, Hub Page (components + state + logic)



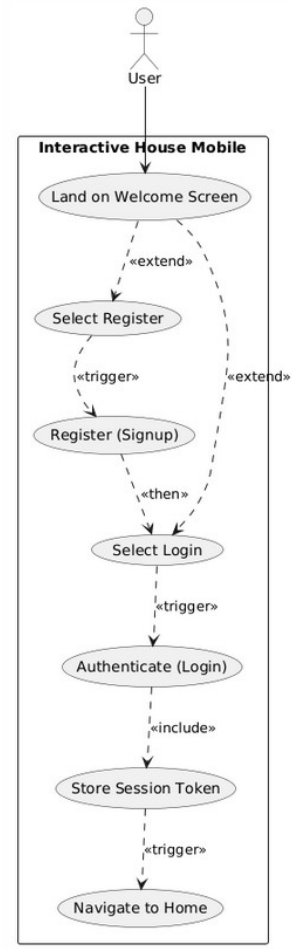
Design Item 3 sequence diagram for the HUB page



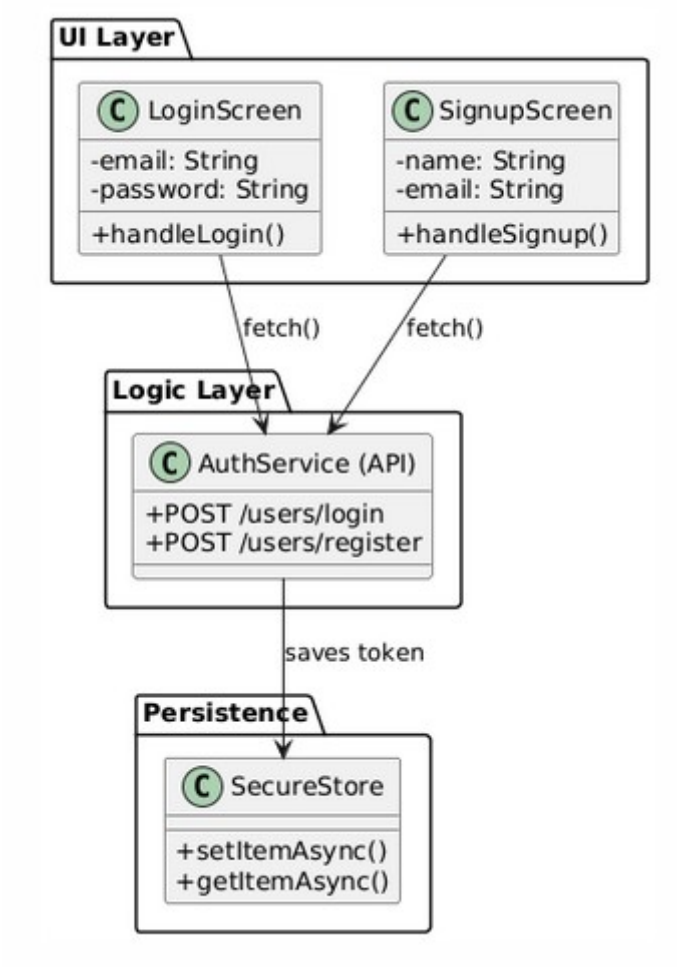
Design Item 4 Hub Page, State Diagram

D5. Mobile Login/Signup - Essential

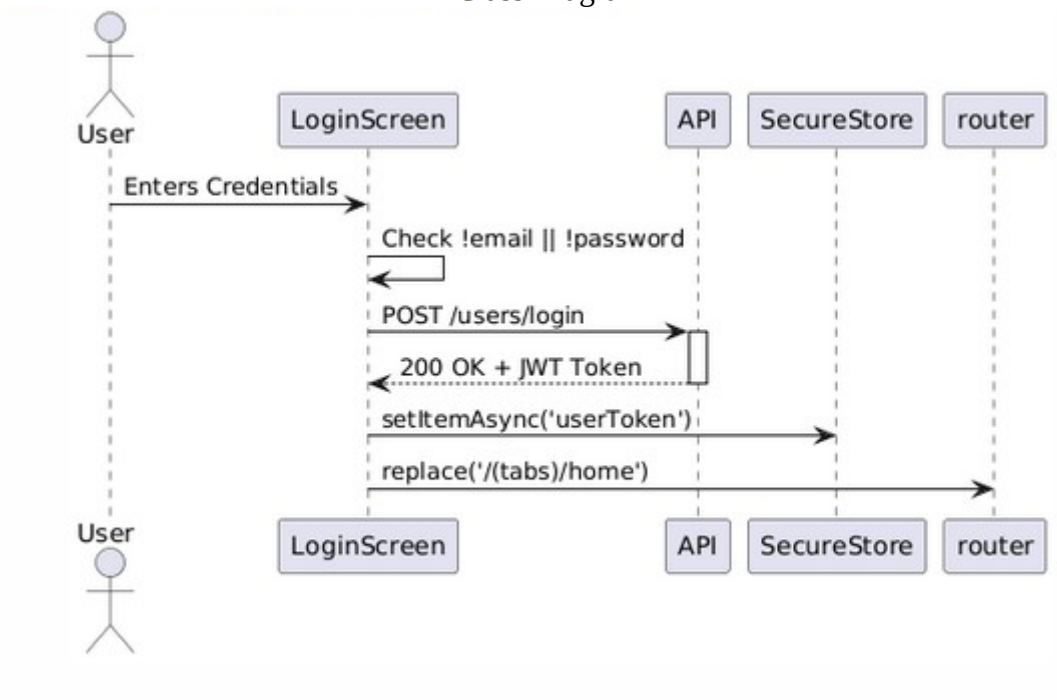
The login/signup module serves as the primary security gateway for the Interactive House Mobile application. This design item manages the transition from an unauthenticated Welcome Screen state to a fully authorized session.
The Use Case diagram illustrates the top-down functional flow, showing how the Welcome Screen extends into Signup and Login actions, ultimately triggering the Navigate to Home use case upon token storage.
The Class diagram specifies the structure of the authentication system, highlighting the data attributes (email, password, name) and core methods (handleLogin(), handleSignup()) that facilitate communication between the UI and the storage utilities.
The Sequence diagram details the chronological communication between the User, API, and SecureStore, specifically showing the logic gate where the system checks for credentials before firing a network request.
The State diagram models the application's behavior at a high level, defining the transitions between LoggedOut and LoggedIn status and the internal logic used to handle "App Start" and session timeouts.



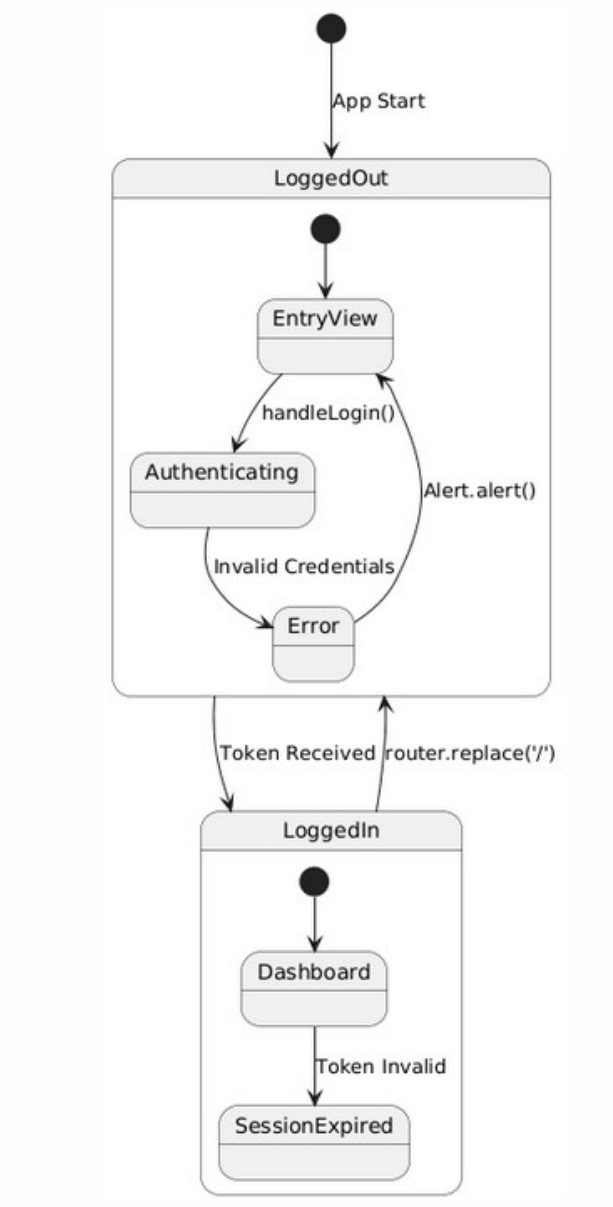
Use Case Diagram



Class Diagram



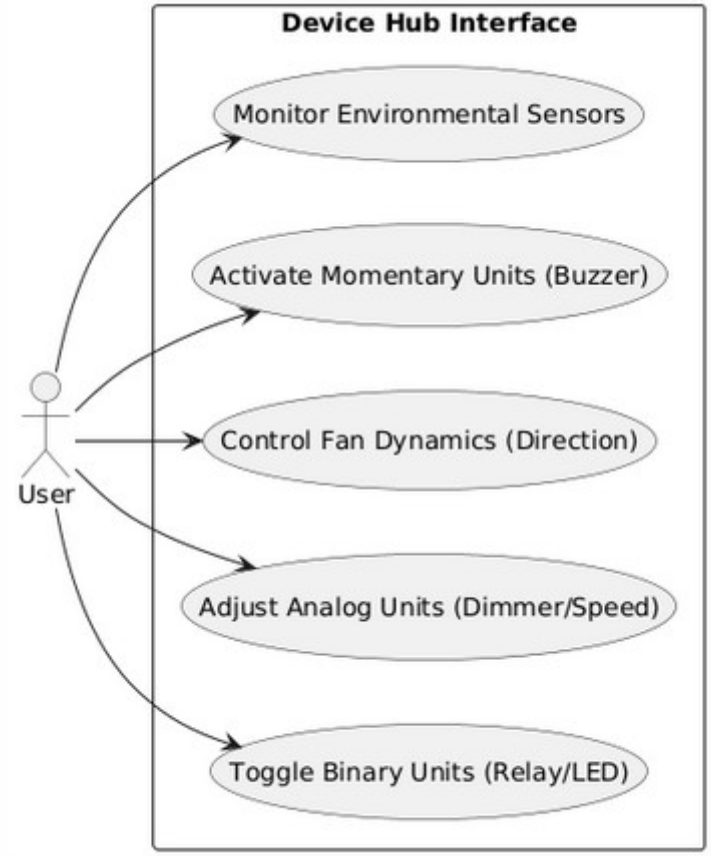
Interaction Sequence Diagram



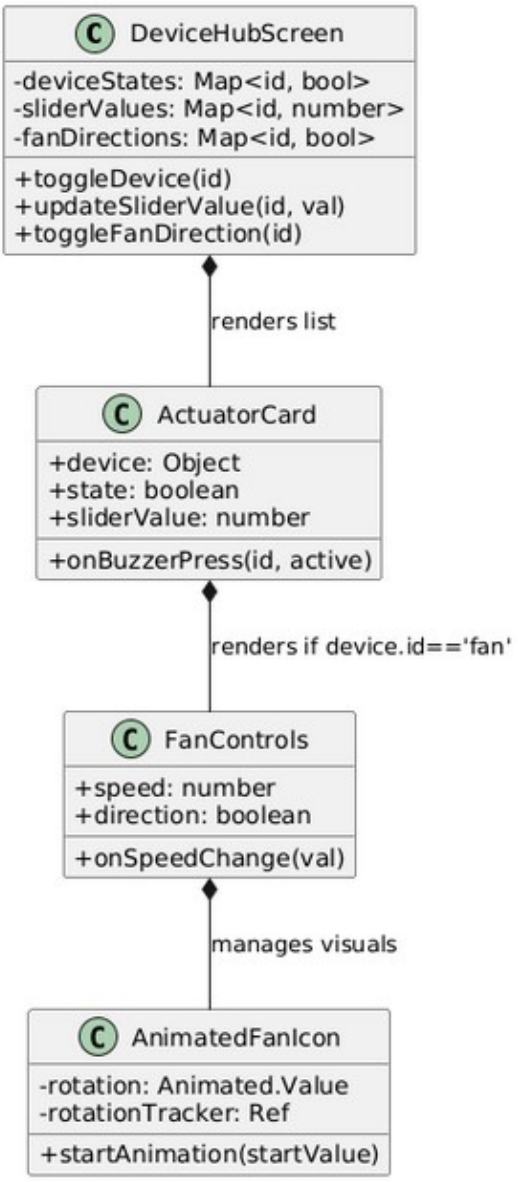
State Diagram

D6 - Mobile Control Devices in Device Hub - Essential

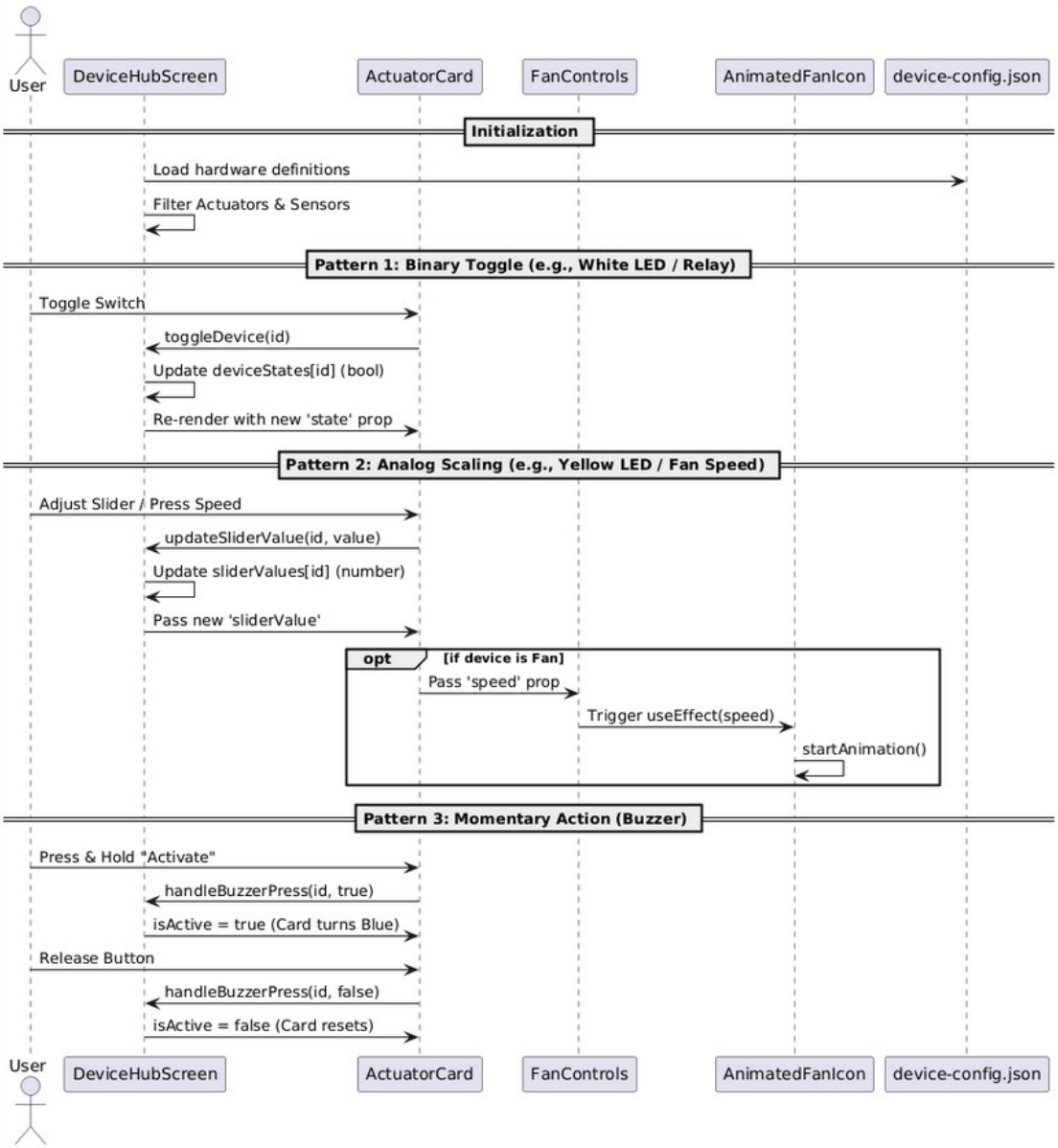
The Device Hub is the core operational component of the mobile, responsible for the real-time control and monitoring of hardware actuators and sensors. The design implements a React Native component hierarchy that translates raw hardware configuration (devices-config.json) into an interactive UI element. This will later be migrated directly through the database connection. However, Json might be good to keep, for mockup data and offline persistence.



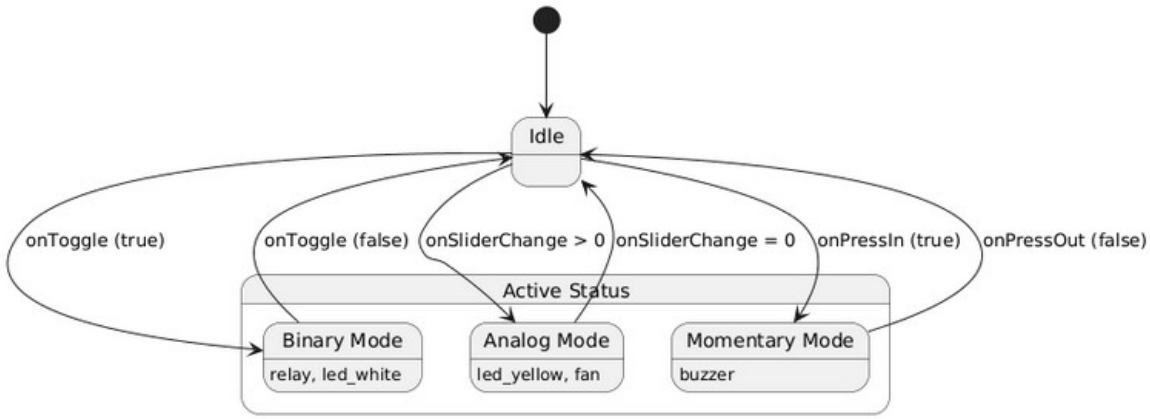
Use Case Diagram



Class Diagram



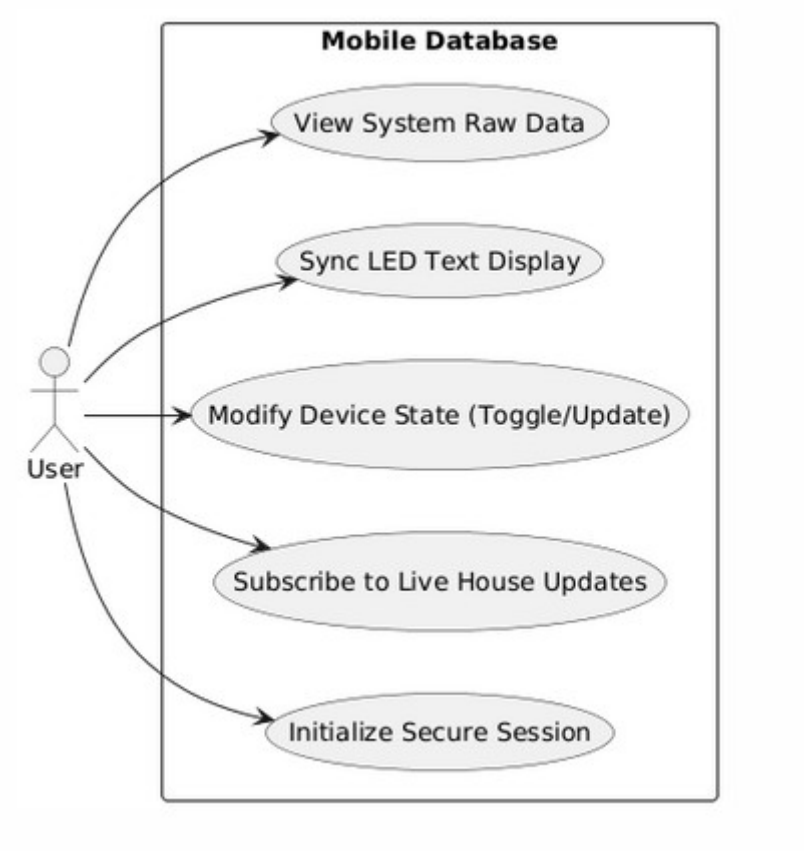
Interaction Sequence Diagram



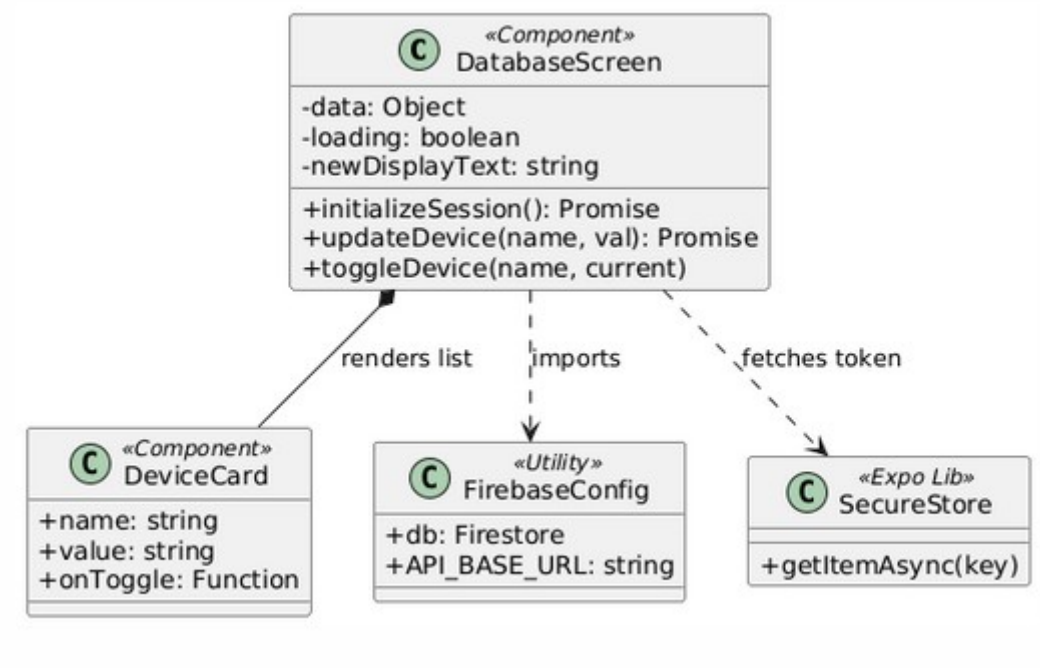
State Diagram

D7 – Mobile Database Connection

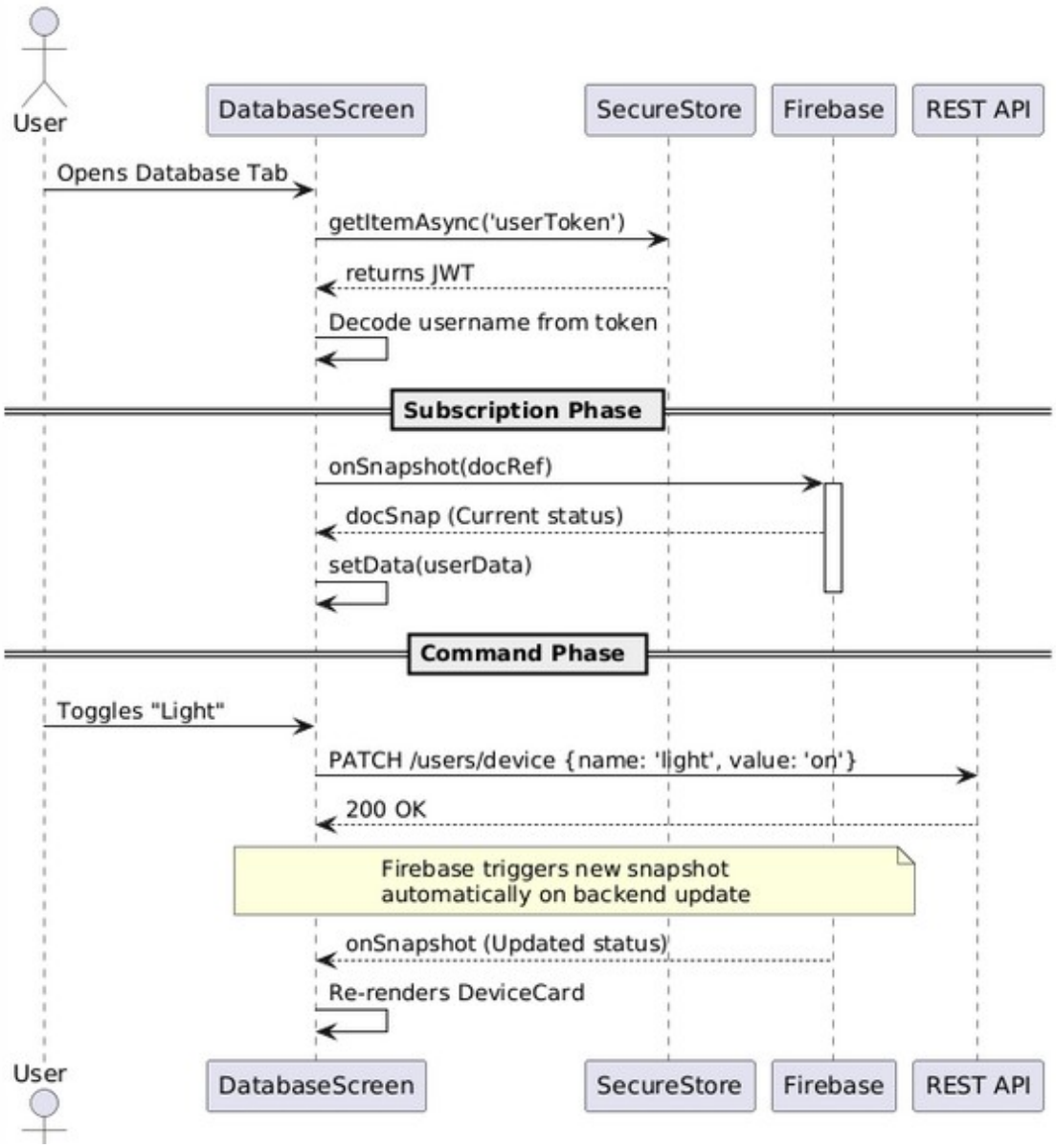
The Database Connection design item ensures data persistence and real-time state synchronization between the mobile client and the smart home hardware. This module bridges the local UI with the cloud backend through a high-performance hybrid architecture.



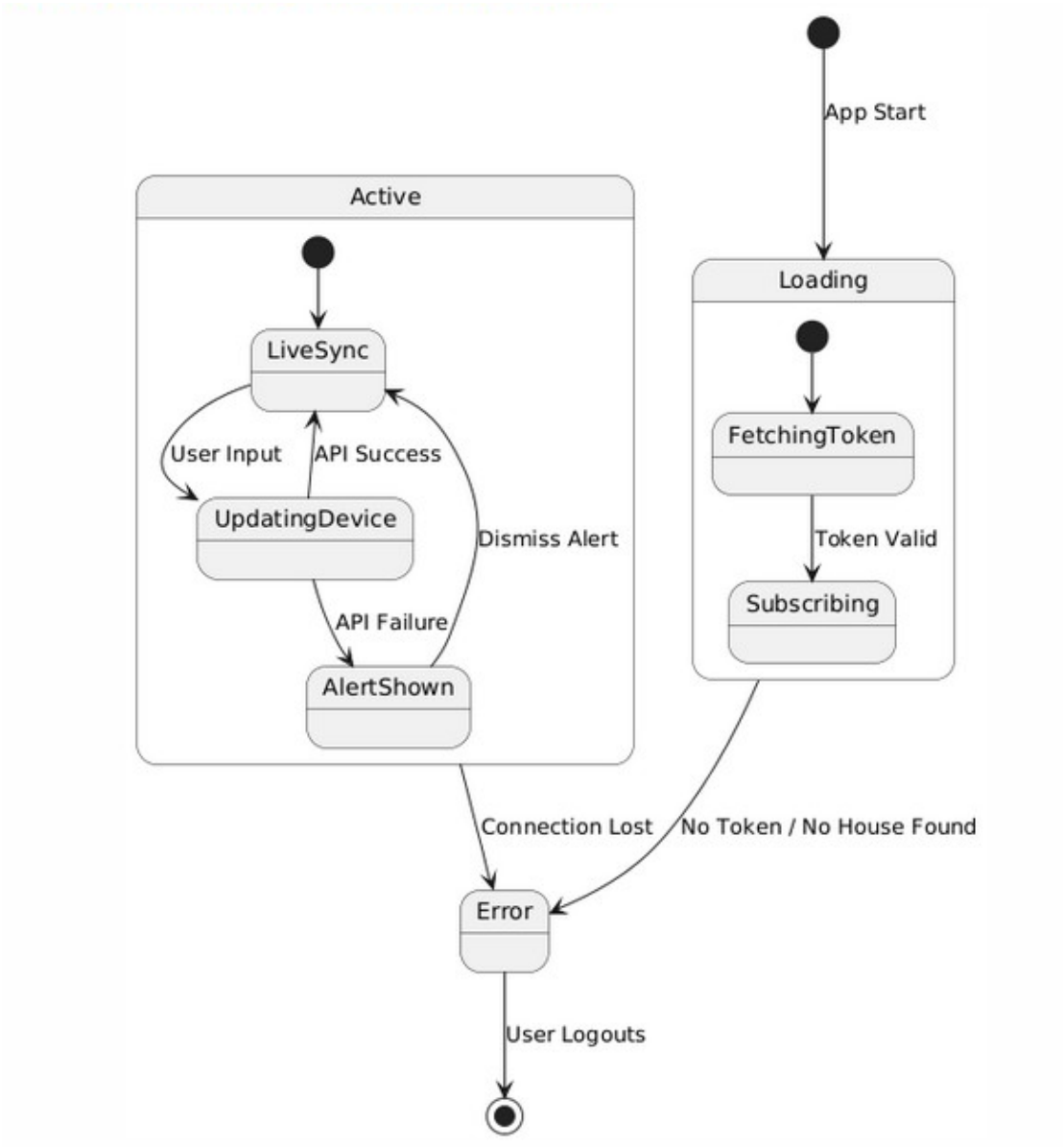
Use Case Diagram



Class Diagram



Interaction Sequence Diagram



State Diagram